

ANX-PR/CL/001-01

GUÍA DE APRENDIZAJE

ASIGNATURA

103001023 - Strategic Management & Entrepreneurship

PLAN DE ESTUDIOS

10AM - Master Universitario En Ingenieria Del Software

CURSO ACADÉMICO Y SEMESTRE

2025/26 - Segundo semestre

Índice

Guía de Aprendizaje

1. Datos descriptivos.....	1
2. Profesorado.....	1
3. Conocimientos previos recomendados.....	2
4. Competencias y resultados de aprendizaje.....	2
5. Descripción de la asignatura y temario.....	4
6. Cronograma.....	9
7. Actividades y criterios de evaluación.....	11
8. Recursos didácticos.....	13
9. Otra información.....	14

1. Datos descriptivos

1.1. Datos de la asignatura

Nombre de la asignatura	103001023 - Strategic Management & Entrepreneurship
No de créditos	4 ECTS
Carácter	Optativa
Curso	Primer curso
Semestre	Segundo semestre
Período de impartición	Febrero-Junio
Idioma de impartición	Inglés/Castellano
Titulación	10AM - Master Universitario en Ingeniería del Software
Centro responsable de la titulación	10 - E.T.S. De Ingenieros Informáticos
Curso académico	2025-26

2. Profesorado

2.1. Profesorado implicado en la docencia

Nombre	Despacho	Correo electrónico	Horario de tutorías *
Claudio Antonio Feijoo Gonzalez (Coordinador/a)		claudio.feijoo@upm.es	- -

* Las horas de tutoría son orientativas y pueden sufrir modificaciones. Se deberá confirmar los horarios de tutorías con el profesorado.

3. Conocimientos previos recomendados

3.1. Asignaturas previas que se recomienda haber cursado

El plan de estudios Master Universitario en Ingeniería del Software no tiene definidas asignaturas previas recomendadas para esta asignatura.

3.2. Otros conocimientos previos recomendados para cursar la asignatura

- Interest in entrepreneurship and / or innovation

4. Competencias y resultados de aprendizaje

4.1. Competencias

CE13 - Tener una visión de los distintos aspectos específicos y emergentes de la ingeniería del software, y profundizar en algunos de ellos

CG1 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio (RD)

CG18 - Capacidad de trabajar y comunicarse también en contextos internacionales

CG3 - Que los estudiantes sepan comunicar sus conclusiones y los conocimientos y razones últimas que las sustentan a públicos especializados y no especializados de un modo claro y sin ambigüedades (RD)

CG9 - Aplicación de los métodos de resolución de problemas más recientes o innovadores y que puedan implicar el uso de otras disciplinas

4.2. Resultados del aprendizaje

RA25 - Communication skills in public SC13, SC14, CG3, CG18 S

RA57 - Exposición y defensa de la solución propuesta de un modo claro y sin ambigüedades ante un público especializado y no especializado.

RA97 - The student is expected to analyse, design and develop an open innovation project in a software-related company

RA26 - Group work skill SC13, SC14, CG17 A

RA95 - The student is expected to understand the basics of strategic management in a software-related company

RA94 - Students are expected to understand and be able to use a number of very practical methods to manage innovative and entrepreneurial projects and performance indicators as part of a company or organization strategy

RA98 - The student is expected to acquire practical knowledge on entrepreneurship techniques such as design thinking, value proposition and lean startup in the context of software-related company

RA22 - Observing capability SC13, SC14, CG10 C

RA21 - Listening capability SC13, SC14, CG10 A

RA23 - Time organization capability SC13, SC14 K

RA24 - Conflict solving capability SC13, SC14, CG18 C

RA27 - Negotiation skill SC13, SC14, CG18 C

RA36 - Posee dotes para liderar el cambio dentro de la organización

RA37 - Está capacitado para introducir mejoras en la organización

5. Descripción de la asignatura y temario

5.1. Descripción de la asignatura

Summary ? Course description

After completing the course, students are expected to understand and be able to use a number of very practical methods to manage innovative and entrepreneurial projects and performance indicators as part of a company or organization strategy.

This subject departs from the idea that the new methods originated in the entrepreneurial arena in the last decade can be extremely useful for managers as they allow to understand the customer better and from the start, rapidly adapting any theoretical project to realistic metrics, acknowledging the fact that innovation and business models are created in a learning process, and that innovation can be ?engineered? to increase the chances of success of any organization.

From a strategic perspective, organizations need to use new approaches to innovation and entrepreneurship to guarantee their competitive advantages in the mid to long term. Open innovation schemes or lean approaches to customers have become key elements of a renewed strategic management.

The course consists of several parts. It starts with an introduction to a revisited strategic management, including economic foundations and the increasingly relevant role of innovation and entrepreneurship. From here a number of cases and projects will be selected for the students to team work on them. All the projects will loosely follow the lean approach to innovation using methodologies such as creative thinking, value creation, ecosystem analysis, business model canvas and lean startup. It will also include practical sessions about other aspects of the strategic management with a view on (open) innovation and entrepreneurship such as the role of organizational structure, technology, communication, IPR and internationalization.

The course will emphasize the practical side of the tools and will introduce related key concepts on the innovation / entrepreneurial ecosystems with a particular focus on software-related companies.

During the course, students will team work in a practical project of their own that will be presented, discussed and evaluated in a final pitch session.

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Objectives and requirements

This is a practical graduate-level course on tools and methodologies for strategic managers based on new developments in entrepreneurship and innovation.

There are three major objectives of this course: (i) to provide students with some conceptual and practical tools to improve their strategic management skills in particular on innovation-related projects and situations; (ii) to provide students with the general rationale and ordered logical steps when approaching innovation from a strategic management perspective; (iii) to provide hands-on knowledge on tools, methods and approaches to tap on innovation and entrepreneurship from the perspective of strategic management.

Students are recommended to enjoy practical interest in innovation and entrepreneurship. Ideally, they should have an innovation strategy for a company or organization of their own, although this is not a strict requisite since during first classes some cases will be provided for the students so they can choose.

Sessions will include work on projects to be developed and discussed during the classes. Therefore, it is highly recommendable to bring a laptop or tablet with basic applications (word processor, presentation software, spreadsheet) and Internet connection. Students are also requested to read the references assigned to each session and to provide results for the assignments.

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Programme & Schedule

NB: This is a tentative programme. The topics are indicative. Program might be modified depending on the interest of students about the subjects covered, on the level and depth of the discussion, and on the availability of the possible invited lecturers.

Session Time Subject

- 1 2h Introduction to the course. Presentation of students (background and motivation). Program, assignments and evaluation. Background: The increasing relevance of innovation and entrepreneurship.
- 2 2h What it is strategy? How to study strategy? The evolution of strategy. Economic foundations of strategy. The role of innovation and entrepreneurship in strategy
- 3 2h Individual work assignment
- 4 2h Economies of scale and scope. Sources of economies of scale and scope. Diseconomies of scale. The learning curve. Diversification.
- 5 2h Value network. Make vs. buy. Integration.
- 6 2h Innovation and Entrepreneurship ecosystem
- 7 2h Industry analysis. Competition and cooperation

- 8 2h The role of innovation. Types of innovation. Open innovation.
- 9 2h Idea market. Team setting and assignment
- 10 2h Entrepreneurship: concepts
- 11 2h Cases in strategic management & entrepreneurship. Design thinking
- 12 2h Strategic positioning. Competitive advantage. Value creation.
- 13 2h DNA of a project.
- 14 2h Value proposition canvas
- 15 2h Testing & experimenting. Customer development. Validated learning. Minimum viable product. Product-market fit
- 16 2h Experiment board.
- 17 2h Business and lean model canvas in a strategic setting. Tools and methodologies for high-risk, high-uncertainty environments
- 18 2h Lean startup and open innovation
- 19 2h Engaging with startups to enhance corporate innovation. Corporation entrepreneurship
- 20 2h Information and value creation
- 21 2h Sustaining competitive advantage
- 22 2h Open science, open software and open standards. User innovation
- 23 2h Preparing the pitch: How to. Pre-mortem analysis. Beyond the pitch
- 24 4h Visits / Guest lectures / webinars / online events from incubators, accelerators and open innovation initiatives
- 25 4h Workshops (pitch) for presentation and discussion of results of workgroups projects

5.2. Temario de la asignatura

1. Introduction to the course. Presentation of students (background and motivation). Program, assignments and evaluation. Background: The increasing relevance of innovation and entrepreneurship
2. What it is strategy? How to study strategy? The evolution of strategy. Economic foundations of strategy. The role of innovation and entrepreneurship in strategy
3. Individual work assignment
4. Economies of scale and scope. Sources of economies of scale and scope. Diseconomies of scale. The learning curve. Diversification
5. Value network. Make vs. buy. Integration
6. Innovation and Entrepreneurship ecosystem
7. Industry analysis. Competition and cooperation
8. The role of innovation. Types of innovation. Open innovation
9. Idea market. Team setting and assignment
10. Entrepreneurship: concepts
11. Cases in strategic management & entrepreneurship. Design thinking
12. Strategic positioning. Competitive advantage. Value creation
13. DNA of a project
14. Value proposition canvas
15. Testing & experimenting. Customer development. Validated learning. Minimum viable product. Product-market fit
16. Experiment board
17. Business and lean model canvas in a strategic setting. Tools and methodologies for high-risk, high-uncertainty environments
18. Lean startup and open innovation
19. Engaging with startups to enhance corporate innovation. Corporation entrepreneurship
20. Information and value creation
21. Sustaining competitive advantage
22. Open science, open software and open standards. User innovation
23. Preparing the pitch: How to. Pre-mortem analysis. Beyond the pitch

24. Visits / Guest lectures / webinars / online events from incubators, accelerators and open innovation initiatives

25. Workshops (pitch) for presentation and discussion of results of workgroups projects

6. Cronograma

6.1. Cronograma de la asignatura *

Sem	Actividad tipo 1	Actividad tipo 2	Tele-enseñanza	Actividades de evaluación
1	Regular session in the classroom Duración: 02:00 LM: Actividad del tipo Lección Magistral			
2	Regular session in the classroom Duración: 02:00 LM: Actividad del tipo Lección Magistral			
3	Individual assignment Duración: 02:00 OT: Otras actividades formativas / Evaluación			Individual assignment TI: Técnica del tipo Trabajo Individual Evaluación Progresiva y Global Presencial Duración: 02:00
4	Regular session in the classroom Duración: 02:00 LM: Actividad del tipo Lección Magistral			
5	Regular session in the classroom Duración: 02:00 LM: Actividad del tipo Lección Magistral			
6	Regular session in the classroom Duración: 02:00 AC: Actividad del tipo Acciones Cooperativas			
7	Teamwork assignment Duración: 02:00 OT: Otras actividades formativas / Evaluación			Teamwork assignment PG: Técnica del tipo Presentación en Grupo Evaluación Progresiva y Global Presencial Duración: 02:00
8	Regular session in the classroom Duración: 02:00 AC: Actividad del tipo Acciones Cooperativas			
9	Regular session in the classroom Duración: 02:00 AC: Actividad del tipo Acciones Cooperativas			
10	Regular session in the classroom Duración: 02:00 AC: Actividad del tipo Acciones Cooperativas			
11	Regular session in the classroom Duración: 02:00 AC: Actividad del tipo Acciones Cooperativas			

12	Regular session in the classroom Duración: 02:00 OT: Otras actividades formativas / Evaluación			
13	Regular session in the classroom Duración: 02:00 AC: Actividad del tipo Acciones Cooperativas			
14	Regular session in the classroom Duración: 01:00 AC: Actividad del tipo Acciones Cooperativas	Participation and contributions during classes Duración: 01:00 OT: Otras actividades formativas / Evaluación		Participation and contributions during classes OT: Otras técnicas evaluativas Evaluación Progresiva y Global Presencial Duración: 01:00
15	Pitch. Teamwork presentation Duración: 02:00 OT: Otras actividades formativas / Evaluación			Pitch. Teamwork presentation PG: Técnica del tipo Presentación en Grupo Evaluación Progresiva y Global Presencial Duración: 02:00
16				
17				Global evaluation is the result of adding the individual + teamwork + participation - Other students need to ask for extraordinary evaluation OT: Otras técnicas evaluativas Evaluación Global No presencial Duración: 00:00

Para el cálculo de los valores totales, se estima que por cada crédito ECTS el alumno dedicará dependiendo del plan de estudios, entre 26 y 27 horas de trabajo presencial y no presencial.

7. Actividades y criterios de evaluación

7.1. Actividades de evaluación de la asignatura

7.1.1. Evaluación (progresiva)

Sem.	Descripción	Modalidad	Tipo	Duración	Peso en la nota	Nota mínima	Competencias evaluadas
3	Individual assignment	TI: Técnica del tipo Trabajo Individual	Presencial	02:00	40%	6 / 10	CG1 CG3 CG18 CE13 CG9
7	Teamwork assignment	PG: Técnica del tipo Presentación en Grupo	Presencial	02:00	20%	5 / 10	CG1 CG9 CG3 CG18 CE13
14	Participation and contributions during classes	OT: Otras técnicas evaluativas	Presencial	01:00	20%	0 / 10	CG1 CG18 CE13 CG3 CG9
15	Pitch. Teamwork presentation	PG: Técnica del tipo Presentación en Grupo	Presencial	02:00	20%	0 / 10	CG3 CG18 CE13 CG1 CG9

7.1.2. Prueba evaluación global

Sem	Descripción	Modalidad	Tipo	Duración	Peso en la nota	Nota mínima	Competencias evaluadas
3	Individual assignment	TI: Técnica del tipo Trabajo Individual	Presencial	02:00	40%	6 / 10	CG1 CG3 CG18 CE13 CG9
7	Teamwork assignment	PG: Técnica del tipo Presentación en Grupo	Presencial	02:00	20%	5 / 10	CG1 CG9 CG3 CG18 CE13

14	Participation and contributions during classes	OT: Otras técnicas evaluativas	Presencial	01:00	20%	0 / 10	CG1 CG18 CE13 CG3 CG9
15	Pitch. Teamwork presentation	PG: Técnica del tipo Presentación en Grupo	Presencial	02:00	20%	0 / 10	CG3 CG18 CE13 CG1 CG9
17	Global evaluation is the result of adding the individual + teamwork + participation - Other students need to ask for extraordinary evaluation	OT: Otras técnicas evaluativas	No Presencial	00:00	%	0 / 10	CG9 CG3 CG18 CE13 CG1

7.1.3. Evaluación convocatoria extraordinaria

Descripción	Modalidad	Tipo	Duración	Peso en la nota	Nota mínima	Competencias evaluadas
Compensation work - Just for the individual assignment. Participation and Teamwork are not possible to be compensated	TI: Técnica del tipo Trabajo Individual	Presencial	02:00	60%	5 / 10	CG3 CG9 CG1 CG18 CE13

7.2. Criterios de evaluación

Evaluation criteria

The evaluation will be based on the individual assignments and teamwork:

? Individual assignment / exam: 40%

? Teamwork (including pitch / online presentation): 40%

? Contributions / participation during classes: 20%

The criteria for evaluation will be:

? Understanding of the concepts explained

? Ability to apply them in an actual strategic setting

? Contributions to discussion

? Thinking outside the box

8. Recursos didácticos

8.1. Recursos didácticos de la asignatura

Nombre	Tipo	Observaciones
Lecture Notes	Recursos web	Notes for each session
[1] Economic Foundations of Strategy, Joseph T. Mahoney, Sage Publications Inc, 2005.	Bibliografía	
[2] Osterwalder, A., Pigneur, Y., & Clark, T. (2010). Business model generation: A handbook for visionaries, game changers, and challengers. Hoboken. NJ: Wiley.	Bibliografía	

[3] Blank, S., & Dorf, B. (2012). The startup owner's manual. K&S; Ranch	Bibliografía	
[4] Ries, E. (2011). The lean startup: How today's entrepreneurs use continuous innovation to create radically successful businesses. Random House LLC.	Bibliografía	
[5] Besanko, D., Dranove, D., Shanley, M., Schaefer, S. (2017). Economics of strategy. Wiley, 7th Ed.	Bibliografía	
[6] Chesbrough, H. (2020). Open innovation results. Going beyond the hype and getting down to business. Oxford: Oxford University Press.	Bibliografía	
[7] M. Mazzucato. ?The value of everything. Making and taking in the global economy?. PublicAffairs, 2018	Bibliografía	

9. Otra información

9.1. Otra información sobre la asignatura

The subject is related to digital governance and tranformation. Therefore has some connection with SDG 16 about promoting peaceful and inclusive societies, providing access to justice for all and building effective, accountable and inclusive institutions at all levels.